

Brian Takita

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SUMMARY

Architected a production real-time multimodal AI avatar pipeline at Presence AI — text, audio, and video lip-sync on NVIDIA H100 GPUs via WebRTC. Now building agent tooling in Rust: git-native correspondence (corky), document-as-interface for persistent AI sessions (agent-doc), and cross-language naming analysis (tagpath). A decade of context engineering — from a domain ontology system (2014) through reactive frameworks to modern agentic systems. Deep expertise in event-driven architecture, reactive programming, and collaborative intelligence workflows where agents and humans work in parallel.

SKILLS

AWS, **AWS CDK**, **AWS Lambda**, **Agentic Systems**, **Agents**, **Angular**, **Azure**, **C#**, **CI/CD**, **Claude**, **Context Engineering**, **Convex**, **CopilotKit**, **Django**, **Docker**, **DuckDB**, **FalkorDB**, **FastAPI**, **Firebase**, **Flask**, **Flutter**, **Go**, **Google Cloud**, **Grafana**, **GraphQL**, **Graphene**, **Heroku**, **Jira**, **MongoDB**, **Next.js**, **Node.js**, **NumPy**, **Pandas**, **PostgreSQL**, **PyTorch**, **Python**, **REST**, **Ray**, **React**, **Ruby on Rails**, **Rust**, **SQLAlchemy**, **SQLite**, **Spec-Driven Development**, **Supabase**, **SvelteJS**, **Terraform**, **TimescaleDB**, **TypeScript**, **Vue.js**, **WebRTC**, **ZeroMQ**, **Zig**, **Zod**, **msgspec**

CAREER HIGHLIGHTS

- **Presence AI** — Architected real-time multimodal AI avatar pipeline — text, audio, and video lip-sync on NVIDIA H100 GPUs via WebRTC
- **Open Source** — corky: git-native correspondence in Rust | agent-doc: document-as-interface for AI agents | RSpec: BDD core contributor | rr: pioneered proxy-based test doubles (now patch in Python)
- **Pivotal Labs** — Mentored developers across 20+ startups. Speaker at RailsConf 2008. Early contributor to Behavior Driven Development
- **TrueCar / honk.com** — Led vehicle review platform development at honk.com (acquired by TrueCar); senior product team at TrueCar
- **Milyoni** — Social Entertainment Platform for Warner Brothers, Lions Gate, Paramount Studios, Universal Music Group
- **Censible** — 8-year ESG analytics platform — Martin Investments analyst portal, ESG Portfolio Rebalancing system
- **Context Engineering** — Domain ontology system (extensible) since 2014 — structuring agent reasoning across domains
- **Reactive Frameworks** — 15-year lineage: backbone-signal → ctx-core → rmemo → relsysjs — applied to modern agentic systems

EXPERIENCE

PRESENCE AI —

LEAD BACKEND AI ENGINEER

2024-04 – 2026-02 (1 yr 10 mos) | Remote

- Built real-time AI avatar pipeline processing video/audio streams with sub-100ms latency
- Designed reactive signal graph architecture for deterministic avatar response timing
- Implemented WebRTC streaming infrastructure for live avatar sessions

- Developed multimodal AI pipeline integrating LLM, TTS, lip-sync models, image generation, and live video/audio/text avatars
- Created context engineering system for agentic workflow orchestration
- Implemented ZeroMQ-based event-driven messaging architecture for real-time avatar pipeline coordination
- Scaled the avatar pipeline with Ray for distributed and real-time parallel compute — orchestrating PyTorch lip-sync inference and NumPy numeric workloads, with msgspec for low-latency message serialization

OPEN SOURCE —

CREATOR / SOLO DEVELOPER

2006 – Present | Remote

- Built corky: git-native correspondence system in Rust syncing email, Telegram, and Slack into scoped markdown mailboxes with per-contact agent context
- Built and deployed Google Cloud Platform applications including a public correspondence kit with OAuth2 + Gmail API integration and private data management tools
- Created agent-doc: document-as-interface for persistent AI conversations with CRDT-based streaming merge for real-time human+agent co-editing
- Built agent-kit: Rust SDK for CLI tools integrating with AI agent loops — session management, tmux pane routing, instruction file discovery
- Created tmux-router: declarative tmux pane routing in Rust for multiplexed agent orchestration across concurrent development contexts
- Built tagpath: cross-language identifier analysis tool in Rust with convention detection, semantic search across 14 tree-sitter grammars
- Built tsift: hybrid code-search engine in Rust (BM25 + vector search) over tree-sitter-parsed codebases, with a provider-neutral graph-store layer — SQLite reference backend plus experimental DuckDB/DuckPGQ and Ladybug backends evaluated through a parity-and-performance promotion gate
- Created lazily-rs/lazily-py/lazily-zig: reactive signals with cross-language ports maintaining consistent semantics across Rust, Python, and Zig
- Created ctx-core: explicit, scalable cross-platform reactive contexts in JavaScript — foundation for rmemo, RelementJS, rappstack
- Created rr (double Ruby): pioneered proxy-based test doubles (mock.proxy, stub.proxy) — pattern now standard as patch in Python's pytest/unittest
- Core contributor to RSpec: the Ruby Behavior Driven Development testing framework
- Created domain ontology system (existence-lang, formerly philosophy, 2014): structuring agent reasoning across domains with extensible scope, context, and resolution — gives agents and humans a consistent vocabulary across code, correspondence, and data contexts

BRIAN TAKITA —

FULL-STACK ENGINEER

2019-01 – present (7+ yrs) | Remote

- MonsterRodHolders: Engineering & operations for a mid-market WooCommerce e-commerce store. Ran database-driven revenue diagnostics — traced a sales dip from the order data to a card-testing bot attack (170+ fraudulent PayPal charges triggering fraud-screening false-declines on real customers) and shipped Cloudflare Turnstile mitigation. Hardened Stripe + PayPal checkout, tuned Cloudflare WAF and Imunify360/ModSecurity rules, instrumented Google Tag Manager, and ran SEO site-structure and traffic root-cause analysis
- Equity Funding Source: Built a full-stack loan deal management platform on Cloudflare Workers + D1 + rappstack/Hono. Admin portal with deal pipeline, lender selection workflow, investor pipeline, and document management; borrower portal with application forms, needs-action dashboard, and multi-file document upload. Stripe billing integration. TypeScript + Bun test suite (741+ tests); 74 D1 migrations deployed to production

- Menu.com: Multi-tenant restaurant ordering and payment platform — consumer web ordering, restaurant back-office, and Cordova mobile app. Stripe tokenization for payment processing. Developed AWS CDK + Lambda serverless functions integrated with AWS Fargate Windows .NET API on ECS
- BestInform: European information platform — pixel-perfect Figma implementation, real-time support messaging. Angular 14, PrimeNG, Firebase
- Guard App: Security service platform — admin portal and mobile apps for patrol management, incident reporting, NFC verification. Flutter, Angular, Firebase, Java Spring Boot
- Drivemond: Taxi and ride-hailing platform for Chicago — passenger and driver mobile apps with real-time matching and GPS tracking. Flutter, AWS
- Galler AI: AI-powered image generation and editing platform — React frontend with Figma design system; cropping, clipping, drawing tools
- SafeSchoolz (Aegis Platform): AI safety platform detecting firearms and alerting authorities — built gun detection model (~85% mAP). Android, .NET, Python/Jupyter, Twilio
- IIPM: Membership and certification platform for Irish pension specialists — member and admin portals. WordPress, PHP, JavaScript/jQuery, Stripe

CENSIBLE —

LEAD PRODUCT DEVELOPER

2016-01 – 2024-04 (8 yrs 4 mos) | Greater New York City Area

- Built Martin Investments analyst portal with ESG rating algorithms, XLSX export handling 10,000+ rows, and SVG/PDF report generation
- Built ESG Portfolio Rebalancing system and ESG analytics platform
- Processed ESG and portfolio datasets with Python and Pandas for the Martin Investments analyst portal — rating computation, data normalization, and report-ready aggregations across large holdings datasets

SOCIALCHORUS —

SENIOR SOFTWARE ENGINEER

2018-02 – 2018-08 (7 mos) | Remote

- Converted legacy Backbone frontend to Svelte & React
- ES5 to ES6/ES2017 migration
- Created best practices for front-end component development

NEO INNOVATION, INC. —

SOFTWARE DEVELOPMENT CONSULTANT

2015-09 – 2015-12 (4 mos) | Greater New York City Area

- Built a document-as-interface application for Tenfore Holdings — a research pipeline where executives email contacts and contextual notes, semantically matched to companies and contacts in Salesforce, then routed to the analyst team

CRYSTALCOMMERCE —

SENIOR SOFTWARE DEVELOPER

2015-01 – 2015-09 (9 mos)

- Migrated production Rails app and re-architected the front end using Node.js, Browserify, Gulp, and Backbone

RUNDAVOO —

PLATFORM ARCHITECT / LEAD FRONT END DEVELOPER

2013-03 – 2014-12 (1 yr 10 mos) | Greater Los Angeles Area

- Led front-end development for Rundavoo, a Google top 75 app in 2014 — responsive, multi-platform (desktop, tablet, mobile, server) application with CORS API integration

MILYONI, INC. —

ARCHITECT / LEAD DEVELOPER

2012-03 – 2013-03 (1 yr 1 mo) | Pleasanton, CA

- Lead developer for Social Entertainment Platform including video, social marketing, gamification, real-time social interaction, and interactive engagements synced with video

- Platform used by Warner Brothers, Lions Gate, Paramount Studios, Universal Music Group, Hearst Media, Starz, Ovation TV, Astro, Focus Features, Funimation, Sundance Now
- Mentored junior & mid-level developers; proponent of flattening communication across the organization
- Implemented event-driven architecture and AWS IAM-based access control for platform services

TRUECAR / HONK.COM —

LEAD DEVELOPER → SENIOR SOFTWARE ENGINEER

2009-08 – 2012-02 (2 yrs 7 mos) | Santa Monica, CA

- Led development of honk.com, a vehicle review and white-label platform used by USAA and Wall Street Journal Autos — acquired by TrueCar
- Partnered on product vision for social car shopping that was key to the acquisition
- Senior member of advanced product team at TrueCar — introduced new products, partner integrations (WSJ.com, USAA.com), and devops automation
- Chef scripts for server and workstation automation; high availability deployment on EC2 and Engine Yard
- Implemented MongoDB-based Car Buyer social media experience logging system for tracking user engagement across the platform

PIVOTAL LABS —

AGILE ENGINEER

2006-01 – 2009-08 (3 yrs 7 mos)

- Mentored developers across 20+ startups, bootstrapping their engineering process, team, and initial product
- Speaker at RailsConf 2008
- Leadership role in common code, practices, and open source (Erector, JS Test Server, Unison)
- Agile Development in the XP tradition: TDD, Pair Programming, Daily Standups, Retrospectives
- Implemented AWS IAM access control and functional reactive programming patterns across client projects

PEER TO PATENT PROJECT —

CONSULTANT

2007 – Present (project-based)

- Key contributor to launching the Peer to Patent pilot application, led by New York Law School — a tool to allow the crowd to review and submit prior art for pending patents

NASPP —

IT ANALYST

2004-02 – 2006-01 (1 yr 11 mos)

- SQL Server DB Admin
- Developed internal CRM with Rails, .NET & MS Access
- Reports for Sales Trends and forecasting

FLEXTRONICS —

WEB APPLICATION DEVELOPER INTERN

2001-05 – 2002-02 (9 mos)

- Developed corporate intranet using Microsoft technologies (ASP, SQL Server)

EDUCATION

University of the Pacific BS, Engineering Physics (1997 – 2002)

CERTIFICATIONS

- Advanced Permaculture Course in Teaching
- Permaculture Design Certificate

LANGUAGES

English: Native